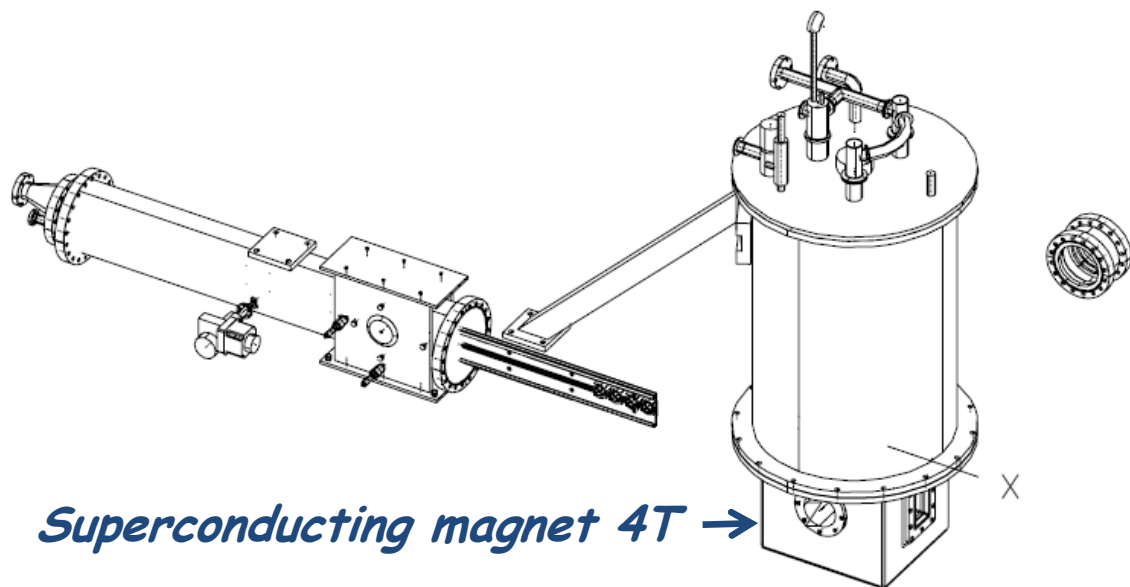


Hall A Møller Polarimeter Update

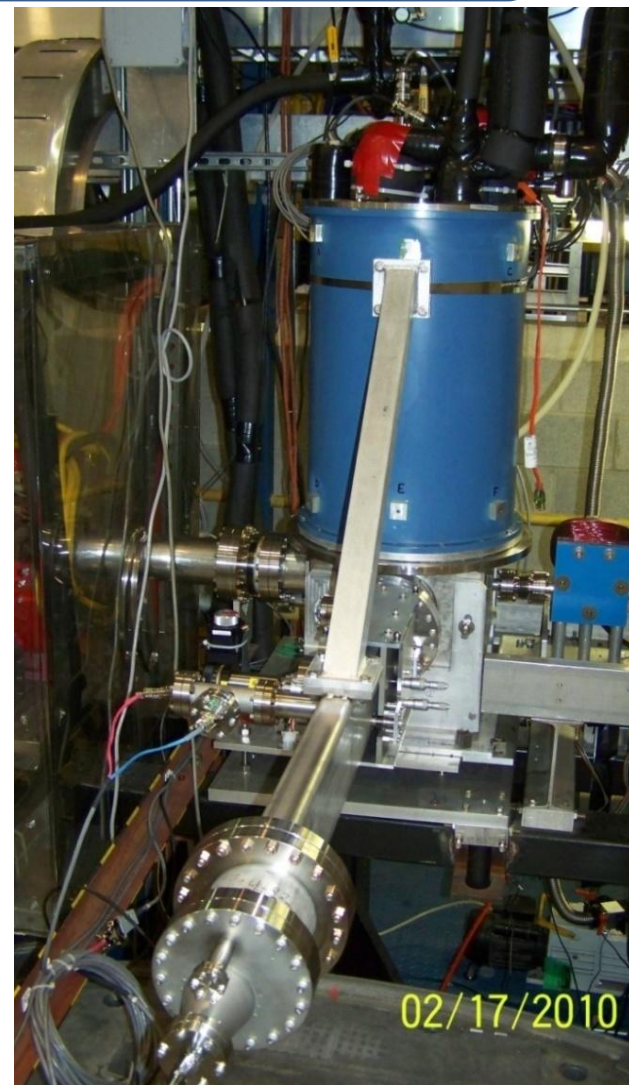
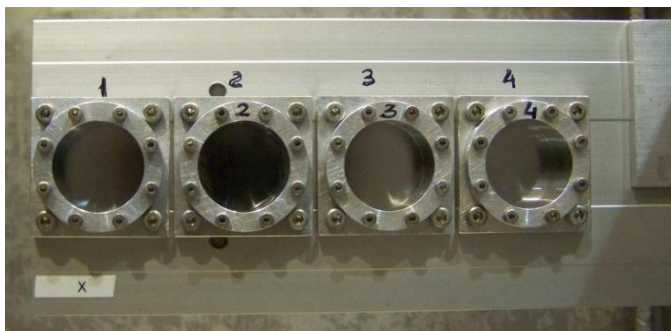
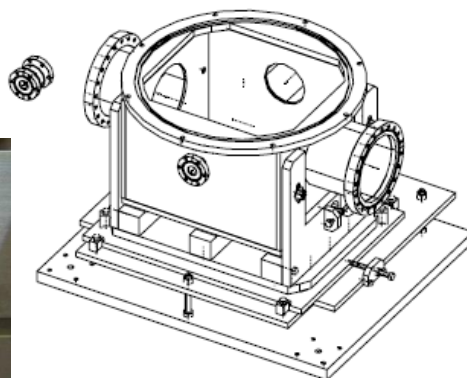
O. Glamazdin

NSC Kharkov Institute of Physics and Technology, Kharkov, Ukraine

“High Field” Target



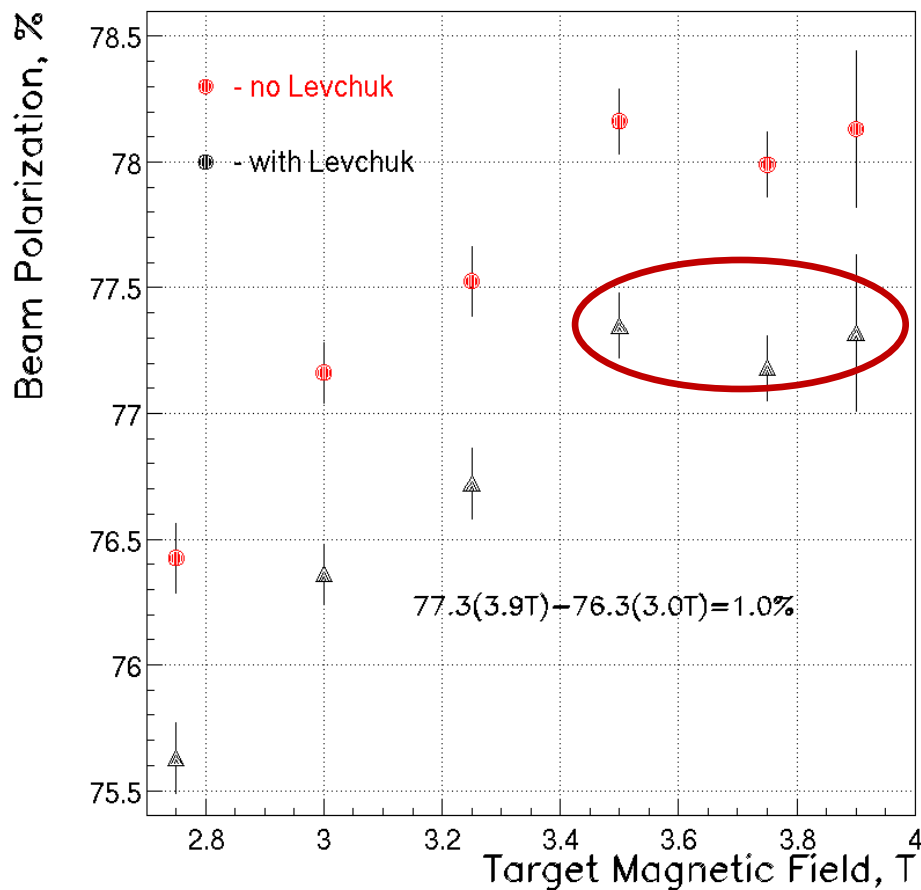
Superconducting magnet 4T →



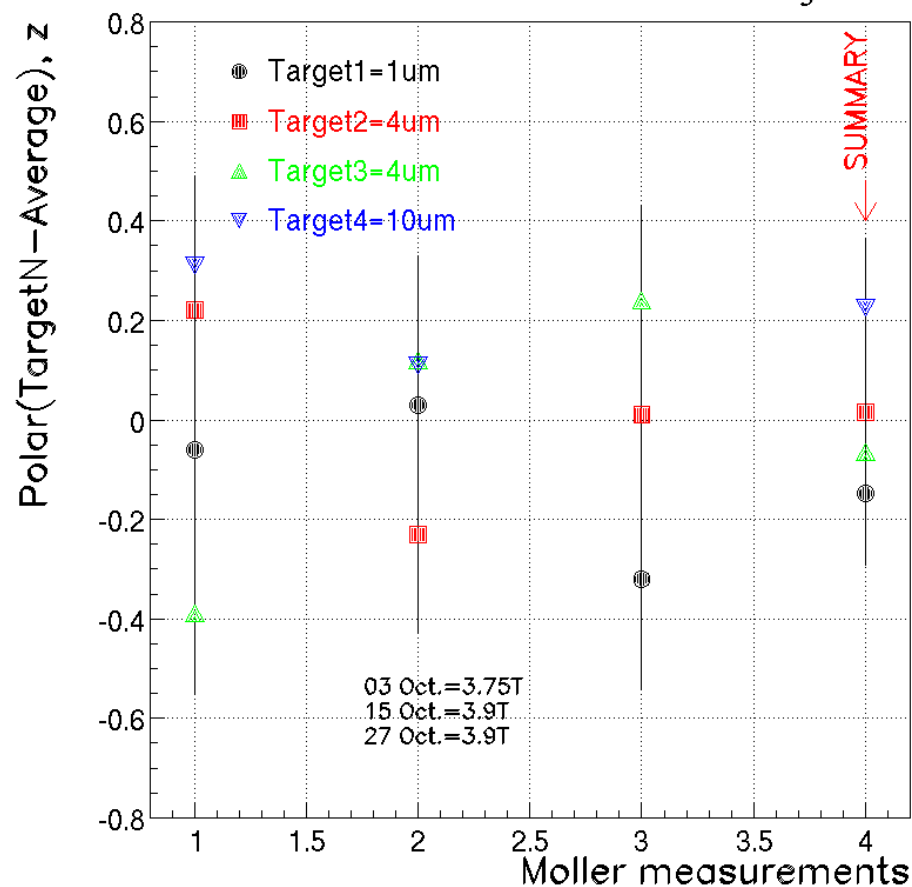
Targets: pure Fe foils 1-1 μ m, 2-4 μ m, 3-4 μ m, 4-10 μ m

“High Field” Target: Results

DVCS Target Saturation Test



DVCS Beam Polarization vs. Moller Target



Møller Polarimeter Systematic Errors

Variable	"Low" field	"High" field 3.9T
<i>Target polarization</i>	1.5%	0.35%
<i>Analyzing power</i>	0.3%	0.3%
<i>Levchuk effect</i>	0.2%	0.3%
<i>Target temperature</i>	0.02%	0.02%
<i>Dead time</i>	0.3%	0.3%
<i>Background</i>	0.3%	0.3%
<i>High beam current</i>	0.2%	0.2%
<i>Others</i>	0.5%	0.5%
<i>Total</i>	1.7%	0.9%

“High Field” Target: Plans

New cryo system

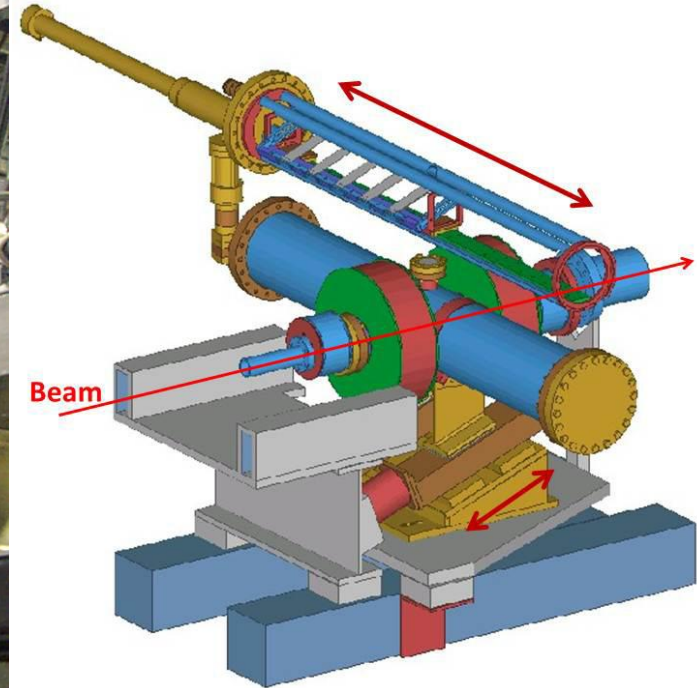
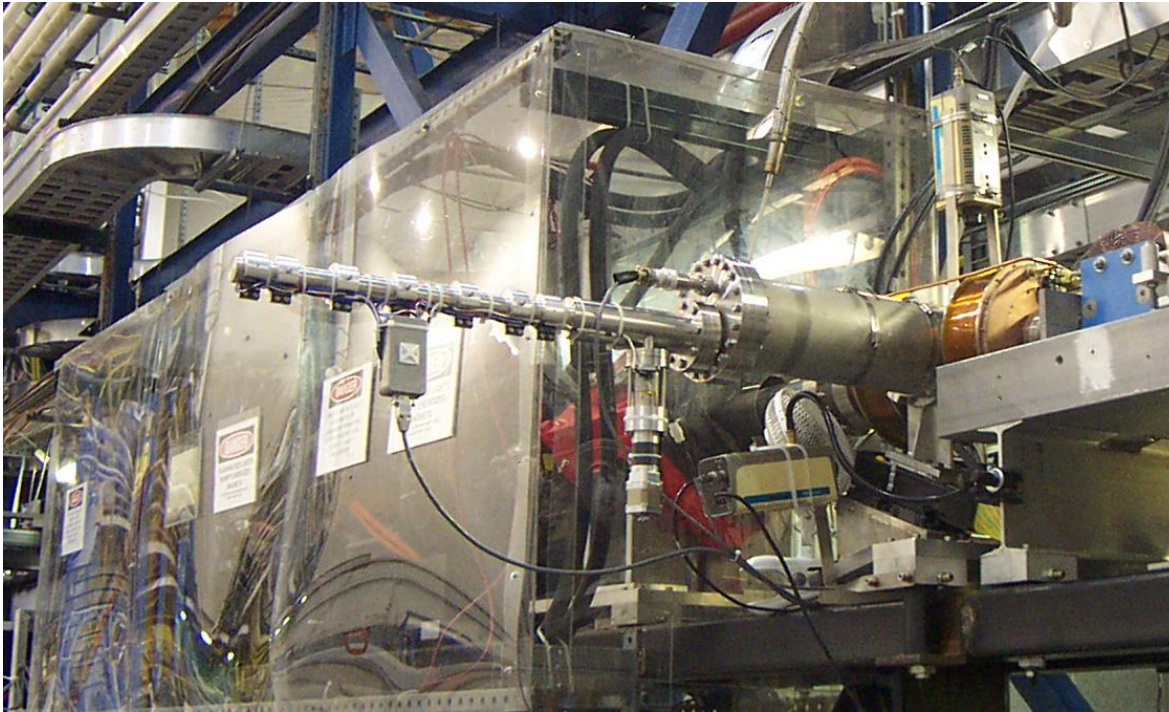
Power supply test

New field mapping

Target holder improvement

New targets and magnetic field alignments

“Low Field” (0.04T) Target



Old target has been re-installed in the beginning of 2011
(cheaper operation, faster data taking at low current)

February 18: was tested on the beam

November–December: target motion, FSD, DAQs tested

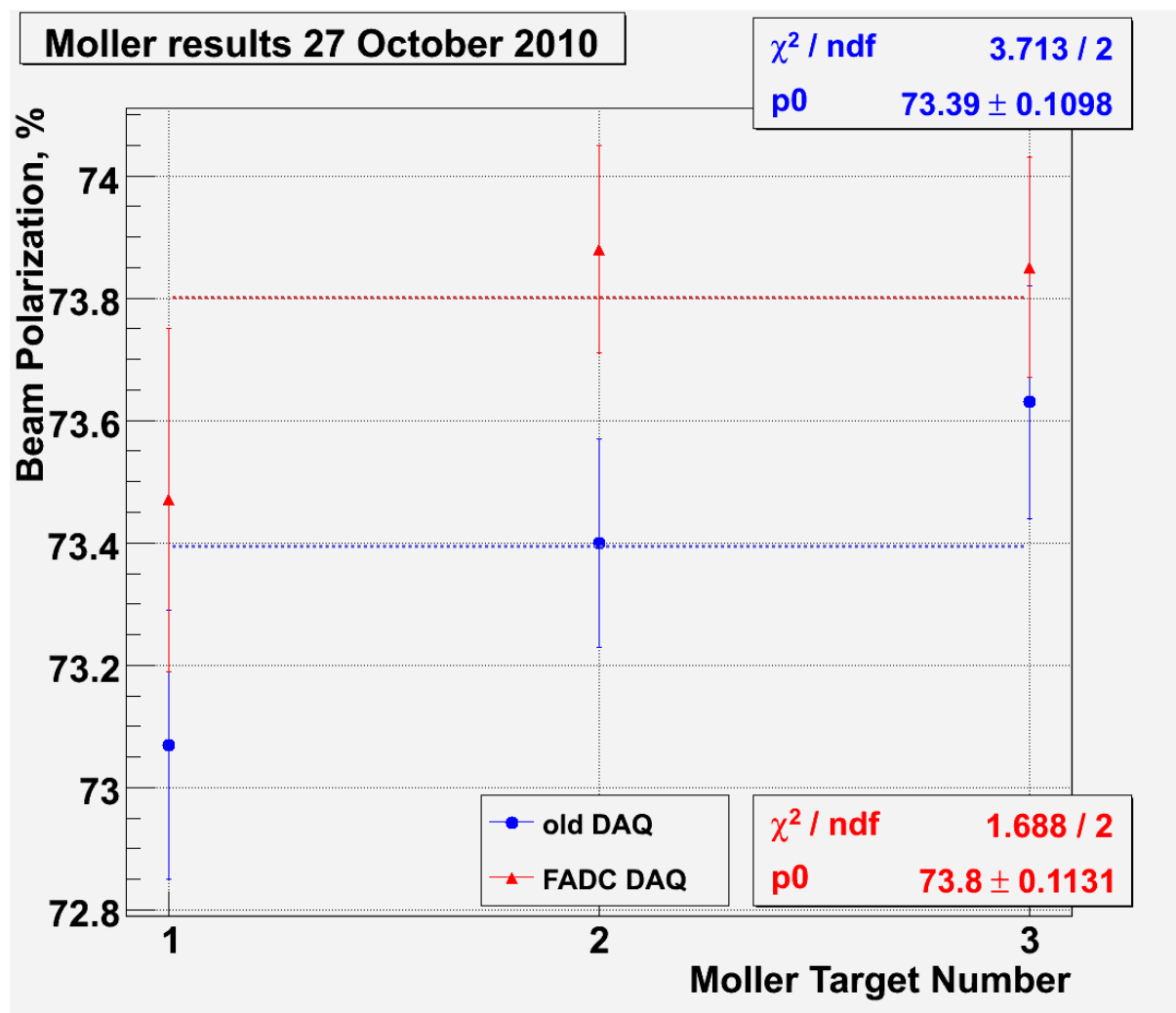
New DAQ based on FADC

The new Moller DAQ based on FADC unit has been used during PREX and DVCS experiments. FADC DAQ was running in parallel with old DAQ.

The main goal of FADC DAQ is to reduce Møller polarimeter systematic error by:

- measuring of the asymmetries using all the event data
- reducing dead time corrections due to FADC pipeline implementation;
- analyzing pileup, analyzing power, Levchuk-effect etc. with "Data Trigger"

Old and New DAQs Comparison



Old DAQ Upgrade

The main goal is to get higher bandwidth (up to 200MHz) .
There are no spares for PLU module etc.

List of CAMAC modules to be replaced:

- PLU type of LeCroy-2365 (bandwidth <75MHz.)
- Discriminator type of Ortec-TD8000 (input rate <150MHz)
- TDC type of LeCroy-2277
- ADC type of LeCroy-2249

List of new modules:

- General Purpose VME Board (as PLU) Caen V1495 200 MHz
- Time to Digital Converter Caen V1190B VME 200 MHz
- Digital to Analog Converter (16ch, 12bit) VMIVME-4140 to remote threshold setting for P/S 708
- Leading Edge Discriminator P/S 708 NIM 300 MHz
- Quad 300MHz Majority Logic P/S 754 NIM 300 MHz
- 32 Channel Multievent QDC Caen V792

11 GeV Upgrade

- Add the 4th quadrupole magnet to cover energy range 0.8÷11GeV
- Add a shielding pipe to the dipole magnet
- Lift up the detector shielding box by 10cm
Møller electrons bending angle $10^\circ \rightarrow 7.3^\circ$ to use the same dipole power supply
- Rearrange the beam line shielding downstream of the dipole for 7.3° angle
- New correctors and BPM downstream of the Møller dipole
- New detector shielding house